Latin American Journal of Central Banking 1 (2020) 100006

Contents lists available at ScienceDirect

Latin American Journal of Central Banking

journal homepage: www.elsevier.com/locate/latcb

Debt limits and fiscal space for some Latin American economies ☆

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ARTICLE INFO

JEL classification:
E27
E44
E61
E62
H62
H63
H68

Keywords:
Debt
Sustainability
Fiscal policy
Primary balance

ABSTRACT

This paper provides evidence on the debt limits and fiscal space for some Latin American emerging economies under the fiscal fatigue approach (Ghosh et al., 2013). We propose the spline technique to estimate each government's reaction function and the sovereign risk premium endogenous to the country's indebtedness. Estimates were made with an unbalanced panel of 13 economies—six from Latin America—for the period 1980–2018. The results vary importantly across the countries. They suggest, for instance, that the Colombian public debt limit would be close to 68% of GDP and that the fiscal space—the gap between the current debt level and the estimated debt limit—would reach 16 percentage points. Among the other Latin American economies, Peru and Chile have the largest fiscal space (50% and 39%, respectively), while Ecuador has the lowest (13%). The findings for Mexico look like those of Colombia.

1. Introduction

Because government debt cannot grow at an excessive rate without a country risking insolvency, a limit understandably exists beyond which a country could enter fiscal unsustainability. Therefore, estimating this debt limit is crucial to prevent public debt from approaching this critical level and to determine the possible fiscal space available. In this paper, we estimate the public debt limit and fiscal space for Latin American emerging economies and some OECD members by using the fiscal fatigue approach (Ghosh et al., 2013). According to this prescription, a country could present symptoms of fiscal fatigue when, given successive debt and interest payments increases, a government's efforts to obtain primary surpluses deteriorate, thus making the government suffer from a fatigue that leads its indebtedness toward a limit.

We deviate from the seminal paper by Ghosh et al. in two key aspects. On the one hand, we employed an alternative technique to estimate the reaction function for emerging economies (cubic splines regression in a panel environment)[1]; on the other hand, we made the sovereign risk premium endogenous to the debt level in order to capture the market's confidence in the countries' capacity to meet their repayments. We employed data from a panel of 13 economies for the period 1980–2018, including six from Latin America, and we also used the World Economic Outlook (WEO; 2019) and International Financial Statistics (IFS; 2019) from the IMF.

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[1] To see other application of spline regression technique, see Almeida et al. (2018).

https://doi.org/10.1016/j.latcb.2020.100006
Received 2 April 2020; Received in revised form 12 June 2020; Accepted 10 August 2020
Available online 9 September 2020
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The results varied importantly across the countries. They suggest, for instance, that the Colombian public debt limit would be close to 68.4% of GDP and that its fiscal space would reach near 16.2 percentage points. The size of this space looks important, but it must be taken with caution due to this indicator's sensitivity to external shocks, as usually faced by Latin American economies. The sovereign risk premium is a central parameter in exercises, so when assessing the sensitivity of the debt limit to 100-basis-point increases and decreases, the Colombian debt limit would decrease by 0.9% of GDP and increase by 2.7% of GDP, respectively.[2] The results also show that Peru and Chile have the greatest fiscal space, at 49.9% and 39.4% respectively, given their low indebtedness and their relatively good macro performance, while Ecuador seems to have the smallest space, at 13.5% of GDP. Unfortunately, the debt limit for Brazil cannot be estimated due to the unusually high positive spread between the real interest rate and the growth rate. In turn, the findings for Mexico are like those of Colombia.

This paper contains four sections, in addition to this brief introduction. Section 2 presents the standard framework for analyzing government debt sustainability, from which the fiscal reaction function arises. Section 3 describes the empirical strategy. Section 4 presents the results, and finally, Section 5 presents the conclusions.

2. The function for the government's reaction and fiscal fatigue

The function for the government's reaction to debt behavior comes from the short-term intertemporal budget constraint,[3] given by

$$b_t = \left(\frac{1+r}{1+g}\right) b_{t-1} - s_t, \tag{1}$$

where b is the debt-to-GDP ratio, s is the primary surplus (as a share of GDP), r is the implicit real interest rate, and g is the real economic growth rate. In Eq. (1), we left aside the seigniorage revenues from the central bank. For simplicity, we also assumed that both the interest rate and economic growth were constant.

Under the simplest short-term standpoint, a fiscal stance can be sustainable if the debt remains constant during the time, which implies that $b_t - b_{t-1} = 0$. Substituting this condition in (1) and clearing s, we find that the required amount of primary surplus for period t must be $s_t = (r-g)b_{t-1}$. This is the starting point retaken by Ghosh et al. (2013) for their debt-sustainability analysis, which establishes a non-linear relationship between a country's primary balance and debt dynamics.

This nonlinear relationship goes back to Bohn (2008, 1998, 1991) and Trehan and Walsh (1991), who argued that a condition for sustainability is that a positive correlation exist between the primary balance and debt. In particular, if the relationship is linear, $s_t = \rho b_{t-1}$, then the debt will decrease at a rate of $(1-\rho)$ per year, so that in n periods, $(1-\rho)^n$, it will be equal to 0, thus fulfilling the non-Ponzi game condition. They also argue that it is plausible to assume non-linearity in this relationship as long as there is a level of debt b_{t-1} and a function f (•) that satisfies $f'(b) > \rho > 0$ for all $b_t \geq b_{t-1}$. This condition ensures that future debt will be lower than considered in the linear specification because it is reduced over time at a faster rate. Thus, the reaction function for the government will be given by:

$$s_t = \mu_t + f\left(b_{t-1}\right), \tag{2}$$

where μ_t captures other determinants of the primary balance (controls). The concept of fiscal fatigue is introduced by establishing a cubic function f (•) of the type $s_t = \mu_t + b_{t-1} + {b_{t-1}}^2 + {b_{t-1}}^3$, as represented in Fig. 1. This shows the expected dynamics of the government's primary balance against different levels of indebtedness (blue curve). The dotted line with a positive slope represents the interest payments at the risk-free rate, while the red line includes a premium and will increase at an increasing rate after a certain level of indebtedness. The latter case is relevant for emerging economies, in which risk premiums are very sensitive to the level of debt and fiscal perspectives, among other key determinants. Debt is arbitrarily subdivided into low, moderately high, and very high levels, and the reaction of the primary balance to debt takes a cubic form.

The relevant segment of debt to be analyzed is when a government's financial liabilities cease to be moderate and begin to increase (b^* onwards). The critical fiscal-fatigue scenario could take place after the level represented by $\hat{b}$. From there, the primary balance deteriorates; therefore, debt reaches a limit $\bar{b}$, which is understood to be undesirable because the government could be facing debt-sustainability problems.

To better understand the fiscal fatigue phenomenon, in Fig. 2, we extracted the relevant area from Fig. 1. The debt limit is found when the fiscal-reaction function (FRF), which also plays a demand-side role,[4] crosses the interest payment with risk premium, which, in turn, plays the supply-side role. Thus, if the primary balance (in blue) is smaller than the interest payment (in red), then as the debt rises, there will be some debt level ($\bar{b}$) above which the debt dynamics become explosive. This happens because the government will enter an insolvency state in which, even with new debt, it can no longer fulfill its capital or its interest payments. Therefore, $\bar{b}$ will become the debt limit level.

[2] As is shown in Section 4.2., the effect of the increase and decrease in the risk premium is asymmetric due to the nonlinearity of the reaction function and the interest payments.

[3] Notably, a stock-flow adjustment term exists in debt increases, which is important in Latin America. Economic Commission for Latin America and the Caribben (ECLAC) (2016) shows that for this region (2008–2015), a change in the public debt of 5.6% of GDP includes a stock-flow effect of 1.7 percentage points of GDP.

[4] Here, we refer to the debt market in which governments offer and investors demand securities. Before $\bar{b}$, the appetite for securities is greater than the supply, which leads to higher prices and a yield reduction until $\bar{b}$ is reached, after which the situation reverts.

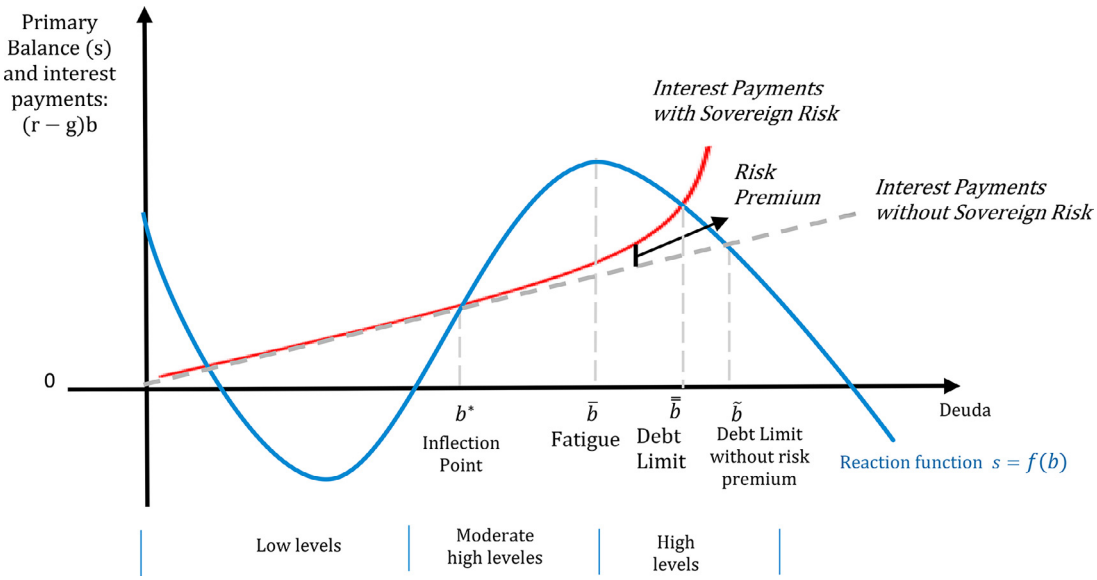


Fig. 1. Fiscal Fatigue: Government Reaction Function and Debt Limit.
Source: Authors' representation based on Ghosh et al. (2013). Fig. 1 shows the government's reaction function (blue line) and the determination of the debt limit (denoted by $\bar{b}$). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

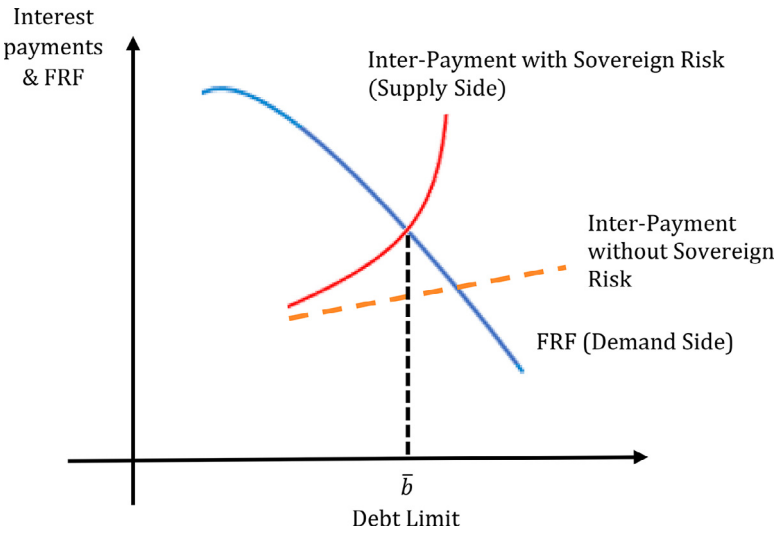


Fig. 2. The Fiscal Fatigue Model (Relevant Area).
Source: Authors representation based on Ghosh et al. (2013).

3. Empirical strategy

Three independent steps were required to estimate the debt limit. First, the government's reaction function was estimated in a panel environment. Second, the spread between the real interest rate and the economic growth was calculated. As mentioned previously, for emerging economies, it is crucial to consider the risk premium in the interest rate. Finally, the debt limit and the corresponding fiscal space were determined using the parameters obtained in the first step and the margins from the second.

3.1. Data

A sample of 13 economies was selected, with the majority representing emerging markets with medium and medium-high incomes. The sample included six Latin American economies: Brazil, Chile, Colombia, Ecuador, Mexico, and Peru. The sample also included a small group of OECD economies, some with high debt levels, to provide greater data variance. To keep the data consistent, we

used the World Economic Outlook (WEO; 2019) and International Financial Statistics (IFS; 2019) of the International Monetary Fund (IMF). Because the coverage of the data differed across countries, an unbalanced data panel was constructed for the period from 1980 to 2018. The fiscal data correspond to the general government, which includes operations of the central, state, and local governments. Table 1 displays the gross public debt for the sample during selected years since 1996. We added the fiscal monitor debt forecasting of October 2019 (IMF, 2019a) (prior to the COVID-19 pandemic) as an indicator of the medium-term outlook.

Some variables suggested by the literature (Galí and Perotti, 2003; Abiad and Ostry, 2005; Bohn, 2008; Mendoza and Ostry, 2008 and Ghosh et al., 2013) were used as controls in estimating the primary balance reaction to the lagged debt. The output gap and government spending gap were calculated using a Hodrick Prescott (HP) filter and are used to control for the effects of business cycles and temporary fluctuations in government outlays. The degree of openness of each economy is assessed as the ratio between exports plus imports to GDP, and inflation was taken as an average of the last three years of consumer prices. Dummy variables were considered for countries with fiscal rules and arrangements with the IMF, with the latter being used as a proxy for international influence on fiscal behavior. Finally, the population dependency ratio was used to consider the country's demographic structure (definition provided by the United Nations; UN).

According to Ghosh et al. (2013), the one econometric issue in this process is the possible endogeneity between debt and country-specific features. A country-fixed effect was included in the regression to control for this source of endogeneity. The real interest rate for Latin American economies corresponds to the deposit interest rate[5] available from the World Bank for the period 2008 to 2018. For the rest of the sample, it corresponds to the debt's effective interest rate,[6] as provided by Article IV of each country, for the period 2008–2018. Real economic growth was taken from the IMF's World Economic Outlook.

3.2. Estimates of the fiscal reaction function

Under the fiscal fatigue approach, the fiscal reaction function for advanced economies is assumed to have a cubic functional form. For emerging markets, however, the standardized fiscal data are unfortunately not very ample, in terms of the time horizon and the number of countries. When observing the fiscal data for our selected sample, we found that a cubic function could imply very strong restrictions that could prevent adequate fit of the data. For this reason, we proposed using a slightly different and more flexible estimation technique known as the cubic spline.[7]

Splines are usually used to estimate the behavior of a pair of variables for an unknown function based on some identified observations. In our case, the observed points were given by the relationship between the primary balance and debt from the sample. Thus, the idea is to approximate the government's fiscal reaction to a cubic function through the use of splines. To do so, in practice, a set of smaller polynomials defined between several debt segments must be constructed that can approach the unknown general function. Debt segments are defined arbitrarily (between k nodes). In our estimate, five symmetrically distributed nodes were considered (debt-to-GDP ratios of 0, 50, 100, 150, and the highest figure observed in the sample).

Rewriting Eq. (2) in a panel environment, we have:

$$s_{i,\,t+1} = \mu_{i,t+1} + f\left(b_{i,t}\right) + \varepsilon_{i,t+1}, \tag{3}$$

where μ captures all the systematic determinants of the primary balances (controls) other than the country's lagged debt, i, and ϵ is an independent and identically distributed innovation bounded in the interval $[-\bar{\varepsilon}, \bar{\varepsilon}]$, given that primary balances are bounded (within only a few points of GDP). In turn, f is the function that will be estimated by applying a cubic B-spline, as follows:

$$f\left(b_{i,t}\right) = \sum_{k=1}^{6} \varphi_{kt}\left(b_{i,t}\right), \tag{4}$$

where φ_{kt} corresponds to the six column vectors calculated with the entire sample debt data and sorted in ascending order. Hence, we finally estimated:

$$s_{i,t+1} = c_i + \sum_{k=1}^{6} \beta_k \varphi_{kt}\left(b_{it}\right) + \Theta\mu_{i,t+1} + \varepsilon_{i,t+1}, \tag{5}$$

where c captures the fixed country effect, β_k corresponds to the estimated parameter for each vector resulting from the spline's application, and Θ is the estimated parameters for the controls. Eq. (5) is estimated by feasible generalized least squares, and autocorrelation of degree one was allowed so that the errors would follow a process:

$$\varepsilon_{i,t+1} = \rho\varepsilon_{i,t} + \nu_{i,t+1} \tag{6}$$

Table 2 shows the results. The model in column 1 includes all of the controls, while column 2 presents the model without those controls resulting in a non-statistical significance or in a correlation between some of them. The signs were as expected, and for the splines' parameters, it was sufficient that one of them had statistical significance in order to obtain significance jointly.[8] Fig. 3 shows the reaction function in blue, which is contrasted with the data (red dots) to examine the goodness of fit to the data.

[5] The rate paid by commercial banks for demand, time, or savings deposits. This rate was used due to restricted data availability.
[6] Defined as interest payments divided by debt stock (excluding guarantees).
[7] For more details on the cubic spline technique go to De Boor (2001).
[8] See details on this matter in De Boor (2001).

Table 1
General Government Gross Debt (Percentage of GDP).

										IMF Outlook Fiscal Mon. Oct. 2019		
	1996	2000	2002	2005	2008	2011	2014	2017	2018	2019	2021	2023
Brazil	N.A.	65.6	78.9	68.7	62.4	61.2	62.3	84.1	87.9	91.6	94.5	95.0
Chile	14.7	13.2	15.2	7.0	4.9	11.1	15.0	23.6	25.6	27.5	29.9	29.6
Colombia	23.4	38.1	47.7	38.6	32.4	35.8	43.3	49.4	52.2	51.0	46.7	42.8
Ecuador	N.A.	N.A.	52.2	34.7	24.2	16.8	27.1	44.6	45.8	49.1	46.6	38.0
Mexico	44.7	40.3	41.9	38.5	42.5	42.9	48.9	54.1	53.6	53.8	54.7	54.9
Peru	N.A.	44.9	45.5	40.4	27.9	23.0	20.6	25.4	26.1	26.9	26.6	25.6
Australia	29.4	19.6	15.1	10.9	11.8	24.1	34.0	41.1	41.4	41.8	41.3	37.9
Belgium	128.0	108.8	104.7	94.7	92.5	102.6	107.5	103.4	102.0	101.0	98.8	96.4
Denmark	68.3	52.4	49.1	37.4	33.3	46.1	44.3	35.5	34.3	33.0	33.8	37.5
Iceland	55.4	37.2	39.1	24.5	66.1	92.0	78.8	43.1	37.6	33.6	29.6	25.4
Japan	101.0	137.9	156.8	176.8	183.4	222.1	236.1	235.0	237.1	237.7	238.4	237.7
South Korea	8.0	16.7	17.0	25.9	26.9	30.2	35.5	37.7	37.9	40.1	46.4	51.3
New Zealand	37.3	30.0	26.4	20.8	19.0	34.7	34.2	31.6	29.8	29.6	30.6	31.4
Memo Item Average: Latin America*	**N.A.**	**46.1**	**58.5**	**50.1**	**46.8**	**47.5**	**50.1**	**62.4**	**69.8**	**71.7**	**70.9**	**69.8**

* Data for Latin America corresponds to the weighted average of individual country data. Data is weighted by annual nominal GDP converted to US dollars.N.A.: Not available.Source: World Economic Outlook Databases for October 2019 and Fiscal Monitor for October 2019, IMF.

Table 2
Spline Estimates of the Fiscal Reaction Function Panel of 13 Countries: 1980–2018.

Dep Variable: Primary Balance	(1)	(2)
Intercept	−16.02***	2.14
Coef. Spline 1	−6.90***	−6.12**
Coef. Spline 2	3.56•	3.11
Coef. Spline 3	−2.82	−4.97
Coef. Spline 4	−4.58	−4.23
Coef. Spline 5	−12.55***	−9.60*
Coef. Spline 6	−10.21***	−4.28•
GAP GDP	0.42***	0.42***
GAP Gov. Spending	−0.29***	−0.31***
IMF Arrangement (Yes = 1)	−0.80	
Fiscal Rule (Yes =1)	2.09*	
Openness	0.018	
Dependent Population	0.34***	
Inflation	−5.97	

Signif. Levels:
*** 0.001,.
** 0.01,.
* 0.05,.
• 0.1, no symbol 1.

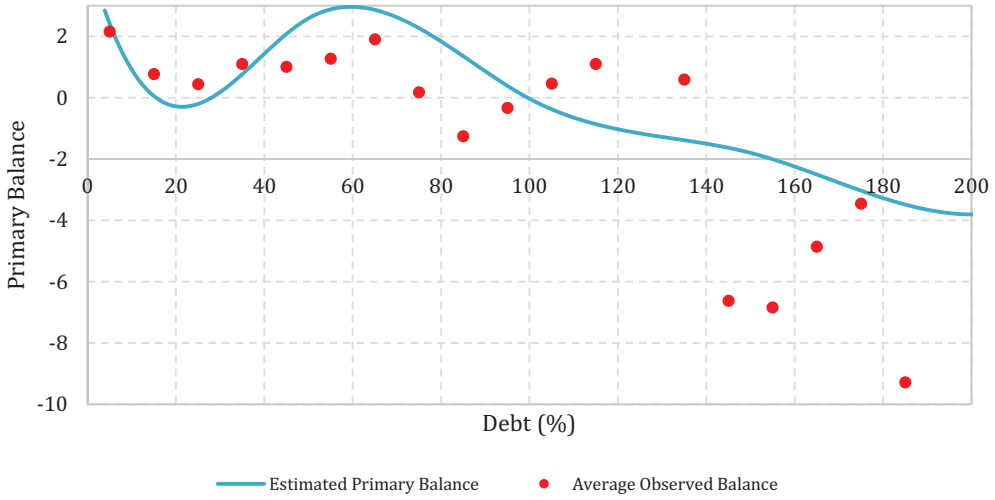


Fig. 3. Reaction Function Estimate vs. Data.
Source: Authors' estimates.

3.3. Spread between the real interest rate and economic growth

In the second step, the spread between the real interest rate and the economic growth rate was estimated for each country in the sample. In the seminal paper by Ghosh et al., there is a low level of risk penalty for advanced economies, which defines the debt limit (noted by $\tilde{b}$, Fig. 1), which in turn is the largest root of the equation $\mu + f(\bar{b}) + \epsilon = (r - g)\bar{b}$. In our case, this property is also satisfied; therefore, the debt limit will arise from:

$$c_i + \sum_{k=1}^{6} \beta_k \varphi_{kt}(\bar{b}_{it}) + \Theta\mu_{i,t} + \boldsymbol{\epsilon_{i,t+1}} = (r_{i,t} - g_{i,t})\bar{b}_{i,t} \tag{7}$$

Nonetheless, because this paper considers the risk premium to be endogenous to the indebtedness level (and indirectly to the probability of default) and, consequently, that the risk premium grows with debt, the interest rate follows a growing and convex path that becomes infinite before $\tilde{b}$.[9] In order to capture this conjecture, which is especially suitable for emerging economies, we identify

[9] For more details, see Lozano et al. (2019).

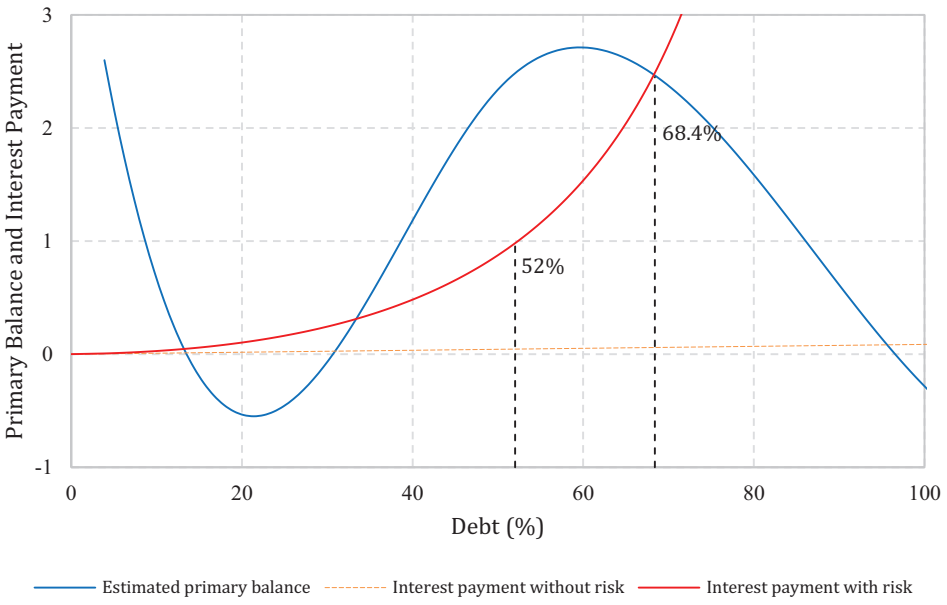


Fig. 4. Debt Limit and Fiscal Space for Colombia.
Source: Authors' estimates.

two components in the real interest rate: the rate without risk ($r^{without\ risk}_{i,t}$) and the risk premium ($rp_{i,t}$). Hence,

$$r_{i,t} = r^{without\ risk}_{i,t} + rp_{i,t} \tag{8}$$

We propose to estimate the risk premium as follows:

$$rp_{i,t} = \begin{cases} \frac{\gamma_i}{(\tilde{b}_i - b_{i,t})^{\beta}} - \delta_i \text{ if } b_{i,t} < \tilde{b}_i \\ \infty \text{ otherwise} \end{cases}, \tag{9}$$

where γ, δ, and β are parameters to be calibrated. Specifically, δ is calibrated in such a way that the risk premium for country i is zero when it has zero debt. In turn, β and γ are calibrated under the following double condition: (i) for each i, the risk premium is equal to that presented in the last year of data, 2018, and (ii) it is consistent with the debt level recorded in that year.

Finally, the debt limit and fiscal space are calculated using the estimated parameters (^) from Eq. (6) and, in addition, by using the country fixed-effect parameter—that is, following $f(\hat{c}_i, \hat{b}_{i,t}, \hat{b}^2_{i,t}, \hat{b}^3_{i,t}, \hat{\mu}_{i,t}) = (\hat{r}_{i,t} - \hat{g}_{i,t})b_{i,t}$. The fiscal space will be given by the difference between the estimated value of the limit debt and the debt level observed in the last available year.

4. Results

4.1. Debt limit and fiscal space for Colombia

To understand the results, in this section, we will briefly analyze findings for one of the Latin American countries, Colombia. Then, we will extend the results to the remaining countries. Fig. 4 shows the results for this particular case. The figures suggest that the limit for Colombia's public debt would be close to 68% of GDP and that its fiscal space would be close to 16 percentage points. The latter number was obtained by considering that Colombia's general government gross debt reached 52% at the end of 2018.

Notably, although the size of this space looks important, it must be taken with caution due to this indicator's sensitivity to external shocks, which Latin American economies usually face. Hence, it is desirable to reverse the increasing trend of the debt recorded in recent years in order to create more fiscal space in the medium term and to preserve access to capital markets. From Fig. 4, one can also infer that the inflection point of the reaction function is around 52% of GDP, which means that risk-rating agencies could be alerted to examine their ratings upon successive increases of public indebtedness above that level.

4.2. Evidence for the full sample

Table 3 shows the results for the total sample. The first three columns display the real interest rate (including the risk premium), the economic growth rate and the spread between these two variables, respectively. These figures correspond to simple averages for the 2008–2018 period. Columns 4 and 5 present the debt (as a share of GDP) and risk premium for 2018, which was the last available

Table 3
Limit of Public Debt and Fiscal Space for Some Emerging and Advanced Economies.

Country	Real Interest Rate* (r) (1)	Economic Growth (g)** (2)	Spread ($r - g$) (Percentages) (3)	General Government Gross Debt 2018 (% GDP) (4)	Risk Premium*** 2018 (b.p.) (5)	Limit of Public Debt (% GDP) (6)	Fiscal Space (% GDP) (7 = 6 − 4)	Fiscal Sensitivity			
								Limit of Public Debt		Fiscal Space	
								−100 b.p. in Risk Premium (8)	+100 b.p. in Risk Premium (9)	−100 b.p. in Risk Premium (10 = 8 − 4)	+100 b.p. in Risk Premium (11 = 9 − 4)
Brazil	9.73	1.57	8.16	87.9	265	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
Chile	4.05	3.12	0.93	25.6	133	65.0	39.4	76.1	61.2	50.5	35.6
Colombia	5.41	3.48	1.93	52.2	184	68.4	16.2	71.1	67.5	18.9	15.3
Ecuador	4.13	3.21	0.92	45.8	642	59.3	13.5	61.1	57.8	15.3	12
Mexico	1.67	2.07	−0.40	53.6	273	69.5	15.9	71.8	67	18.2	13.4
Peru	2.79	3.99	−1.20	26.1	148	76.0	49.9	91.9	66.9	65.8	40.8
Australia	1.88	2.76	−0.88	41.4	−22	99.5	58.1	99.5	84.6	58.1	43.2
Belgium	1.23	1.21	0.02	102.0	39	150.0	48.0	151.8	133	49.8	31
South Korea	2.10	3.37	−1.27	37.9	−40	166.2	128.3	166.2	104.2	128.3	66.3
Denmark	2.31	0.83	1.48	34.3	6	76.1	41.8	76.1	61.8	41.8	27.5
Iceland	4.27	2.59	1.68	37.6	−106	74.0	36.4	74	81	36.4	43.4
Japan	1.03	0.64	0.39	237.1	−33	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
New Zealand	2.87	2.44	0.43	29.8	−15	134.0	104.2	134.5	88.1	104.7	58.3

* For the Latin American (LA) economies (excluding Peru), *r* corresponds to the deposit interest rates retrieved from the World Bank (2008–2018). For the OECD economies and Peru, *r* corresponds to the real interest rate paid by the debt (2008–2018). The source differences are due to data availability restrictions.

** For the LA economies (excluding Peru), the period considered was 2008–2018. The period considered was 2007–2018 for the OECD economies and 2011–2018 for Peru.

*** For the LA economies, the risk premium is the 2018 EMBI. For Australia, Iceland, South Korea, and New Zealand, it is the difference between the yield on a long-term bond issued by that government and one issued by the U.S. Treasury. Finally, for Belgium, Denmark, and Japan, the same spread was used but with a German bond.N.A.: Not available.Source: Authors' estimates.

year. The sixth and seventh columns present the main results—that is, the limit of public debt and its corresponding fiscal space, respectively. Finally, we added columns 8 through 11 to assess fiscal sensitivity—that is, to see how much the debt limit and fiscal space would change when the risk premium is increased or decreased by 100 b.p.

Regarding the 2018 government debt for the region's economies, Colombia stood below the average level among the Latin American countries (52.2% vs. 69.8% of GDP in 2018, respectively). Chile and Peru recorded the lowest levels (25.6% and 26.1%, respectively), while Brazil had the highest (87.9%). The recent debt level and the spread between interest rates and economic growth are the key determinants for a country's debt limit and fiscal space.

With respect to fiscal space, Peru (50% of GDP) and Chile (39%) seemed to have the highest space, while Ecuador (13%) had the lowest. The fiscal space for Mexico (16%), in turn, looked like Colombia's. Unfortunately, the debt limit for Brazil, and therefore its fiscal space, could not be estimated due to the unusual high spread between the country's real interest rate and economic growth. To understand why, the spread shown in Fig. 1 (the slope shown in red) was so high that it failed to intersect with the reaction function.

The rest of the sample, which included some emerging and other OECD economies with relatively high debt, generally had more fiscal space. There were some cases in the whole sample—Peru, Mexico, South Korea, and Australia—with negative spreads between interest rates and economic growth. These cases deserve attention because they are not the only ones that currently exhibit negative spreads in the international context.

Under the principle of fiscal sustainability (and debt limit estimation), the spread between the real interest rate and economic growth must be positive for the long-term debt path to be dynamically stable. The positive spread must also satisfy the transversality condition and thus avoid the so-called non–Ponzi game schemes. Nonetheless, the debt dynamics will theoretically become unstable given cases of negative spreads, such as those recorded in recent years—especially due to low international interest rates. Practically, this implies that governments could tolerate higher levels of indebtedness, at least temporarily (Blanchard and Summers, 2019), but with some potential long-term costs (Boskin, 2020).

Finally, we present an exercise to assess the debt limit's sensitivity to changes in the risk premium, which is one of the most relevant exogenous variables to capture macroeconomic concerns. A 100-basis-point (b.p.) increase and decrease in the risk premium are considered with respect to the value recorded in 2018, on average. The findings suggest that in Colombia, for instance, the debt limit could increase by 2.7% of GDP (from 68.4% to 71.1%) in the "good times" (i.e., when the risk premium could fall by 100 b.p.) or be reduced by 0.9%, up to 67.5% of GDP, in "difficult times" (i.e., when the premium increases by 100 b.p., as in 2009, as result of the global financial crisis). We also show that the most sensible country in the Latin American sample was Peru, with the described shocks increasing its debt limit by 15.9% of GDP and reducing it by 9.1% of GDP. The asymmetric effects are because of the nonlinearity of both the reaction function and the interest payments.

5. Conclusions

This paper provided estimates on the limits of public debt and fiscal space in some emerging Latin American economies using the fiscal fatigue approach (Ghosh et al., 2013). According to this theoretical prescription, a country could present these symptoms when, in the face of successive increases in debt and interest payments, a government's efforts to sustain a path of primary fiscal surpluses deteriorate. Consequently, fiscal efforts suffer a state of fatigue, which drives indebtedness to a limit.

In practice, the debt limit must be understood only as a reference indicator whose estimation is valuable at least for two reasons: to prevent government liabilities from approaching that level and to determine the potential fiscal space available for the government. The resulting fiscal space should be taken with caution due to its sensitivity to macroeconomic shocks.

Methodologically, we deviated from Ghosh et al. (2013) in two key aspects. On the one hand, we proposed an alternative technique with which to estimate the government reaction function for emerging economies (cubic splines). On the other hand, we made the sovereign risk premium endogenous to the debt level in order to capture market confidence in the countries' capacity to meet their repayments. We employed data from an unbalanced panel of 13 economies for the period 1980–2018, including six from Latin America, and used the WEO (IMF, 2019b) and IFS (IMF, 2019c) databases.

The results for the Latin American economies varied in important ways. They suggest that the public debt limit would vary between 59% and 76% of GDP and the fiscal space would vary between 13 and 50 percentage points. The fiscal space looks apparently ample in size; however, these results must be taken with caution given this indicator's high sensitivity to external shocks, which this region usually faces. We performed a sensitivity analysis of the debt limit and fiscal space to the risk premium changes for the entire sample, with this being one of the most relevant exogenous variables to capture macroeconomic challenges. In particular, in the regional context, we found that Peru (50% of GDP) and Chile (39%) seemed to have the highest fiscal space, while Ecuador (13%) had the lowest. The fiscal space for both Mexico and Colombia was close to 16%. Unfortunately, the debt limit for Brazil, and therefore its fiscal space, could not be estimated due to the unusual high spread between its real interest rate and economic growth.

We also paid attention to cases with negative spreads between interest rates and economic growth, due in particular to the current conjuncture of low interest rates in the international context. Even though, from a long-term perspective, this would imply that a debt path is dynamically unstable, in practice, this could lead governments to tolerate higher levels of indebtedness, at least temporarily, even though doing so could have some long-term costs.

To conclude, it is important to remark that our estimates were made from a long-run perspective and considering times of economic normality (not extreme circumstances), which is no longer the case due to the COVID-19 shock. This crisis that the world is now facing will cause lower economic growth, an increase in interest rates, and a deterioration in fiscal accounts, all of which can lead countries to have a lower fiscal space than those presented here.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.latcb.2020.100006.

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